

Training & Development Portfolio

Automotive Systems Fundamentals

A portfolio project demonstrating an end-to-end approach to instructional and learning program design.

Prepared as portfolio evidence for a Training & Development Specialist application.

1. About the Designer

I have a background in Industrial Technology Education, with practical exposure to automotive systems and technical education. My experience developing lesson plans, learning objectives, instructional content, practical activities, and assessments has shaped my approach to designing structured learning experiences. This portfolio applies that foundation to a complete sample learning program.

2. Program Overview

This portfolio project demonstrates my approach to transforming a technical subject area into a structured learning experience, from defining measurable objectives through content development, learning activities, practical application, assessment, feedback, and evaluation.

Program title	Automotive Systems Fundamentals
Target audience	Beginner-level vocational and technical learners
Delivery format	Blended learning — instructor-led sessions, demonstrations, discussions, and practical activities
Suggested duration	5 modules / approximately 15 learning sessions
Program goal	Build foundational understanding of major automotive systems while developing basic practical inspection, problem-solving, and diagnostic skills.

3. Program Learning Objectives

- Identify the purpose and major components of common automotive systems.
- Explain, in simple technical terms, how major vehicle systems operate and interact.
- Apply appropriate workshop safety procedures and basic technical vocabulary.
- Carry out basic inspection and fault-identification activities using a structured process.
- Demonstrate learning through quizzes, practical tasks, assignments, and a final assessment.

4. Program Structure

Module	Focus	Learning activities	Assessment	Expected outcome
1	Automotive Fundamentals & Safety	Interactive instruction; component identification; safety demonstration; discussion	Short quiz; safety checklist	Explain basic systems and follow workshop safety procedures.
2	Engine Systems	Diagrams; demonstrations; guided inspection; practical identification	Component task; quiz	Identify key engine components and explain their functions.
3	Automotive Electrical Systems	Circuit demonstration; guided troubleshooting; practical exercises	Circuit task; troubleshooting questions	Understand basic electrical flow and follow a simple fault-finding process.
4	Transmission, Braking & Steering	System diagrams; case studies; demonstrations; inspection	Case-study worksheet; inspection checklist	Explain system purpose and perform basic inspection steps.
5	Diagnosis, Maintenance & Capstone	Fault scenarios; group problem-solving; guided diagnosis; practical task	Final quiz; practical assessment; capstone	Apply a structured approach to basic diagnosis and maintenance decisions.

5. Instructional Design Approach

Each module follows a consistent learning pathway:

- Activate prior knowledge.
- Introduce concepts using clear explanations and visual materials.
- Demonstrate the skill or worked example.
- Provide guided practice.
- Move learners toward independent application.
- Assess understanding and provide feedback.

- Recap learning and connect it to the next topic.

6. Sample Module Build — Engine Systems

Learning outcomes

- Identify the spark plug, piston, cylinder, and crankshaft.
- State the basic function of each component.
- Describe at an introductory level how combustion produces mechanical movement.
- Use a simple checklist to inspect selected engine components safely.

Learning activities

- Examine a labeled engine diagram and identify familiar components.
- Use a real or model engine for component demonstration where available.
- Observe a basic visual inspection demonstration.
- Complete a guided component-identification exercise.
- Work through a simple engine fault scenario.

Formative assessment

1. What is the function of a piston?
2. What does a spark plug do in a petrol engine?
3. What is the role of the crankshaft?
4. Name two safety precautions to observe before inspecting an engine.

7. Assessment Strategy & Rubric

Assessment is embedded throughout the learning journey and combines knowledge checks with practical application.

Criterion	Excellent	Competent	Needs improvement
Technical understanding	Accurately explains concepts and relationships	Explains main concepts with minor gaps	Shows major gaps or confusion
Practical execution	Completes task safely and independently	Completes task with limited guidance	Requires substantial guidance
Problem solving	Uses a logical diagnostic process	Follows a basic troubleshooting sequence	Approach is inconsistent
Communication	Clear, structured and technically appropriate	Generally clear	Unclear or incomplete

8. Capstone / Final Assessment

Learners receive a simple vehicle fault scenario. They identify the relevant system, state likely causes, propose inspection steps, observe safety requirements, and communicate their recommended next action. The facilitator scores the task using the structured rubric.

9. Completion Criteria

- Complete the required learning sessions.
- Meet the minimum score for the final knowledge assessment.
- Complete required practical activities safely.
- Demonstrate understanding of core automotive systems.
- Demonstrate a basic structured approach to diagnosis and maintenance.

10. Design Rationale

The program progresses from foundational knowledge to guided practice and independent application. Learning objectives are aligned with activities and assessment methods. The design uses explanation, visual material, demonstrations, discussion, hands-on practice, feedback, and evaluation to support different learning needs.

11. Skills Demonstrated

- Learning objective development
- Instructional program design
- Curriculum structuring
- Educational content development
- Learning activity design
- Assessment and quiz design
- Rubric development
- Practical learning design
- Learner evaluation
- Technical communication

12. Portfolio Note

This is a sample portfolio project designed to demonstrate instructional and learning program design capability. It is not presented as a claim of prior corporate training delivery. It reflects an approach grounded in technical education and subject-matter knowledge.